

Electronics industry: Connecting the dots

It is obvious that there is a lot of electronics and information technology involved, right from the computer-aided design tools used for designing green homes, their wiring systems, *et al* to a plethora of gadgets, sensors, control mechanisms and energy management tools used. What is more, there is a growing demand for these products, given that the green building trend is catching on.

"The electronics industry will greatly benefit from making products that will help drive the energy costs lower. Whether it is active power management in a cellphone or maximum power-point tracking in a solar panel, there is energy to be saved. This will require innovation both at the chip and system level. There are alliances being established between solar panel makers, sensor makers, standards committees, and the electronics industry to drive energy-efficient solutions that will benefit all. The Bureau of Energy Efficiency (BEE) is working with a set of corporations to help drive the standards. This will bring about the required awareness," says Ananthaswamy.

Once awareness and inclination emerge, not just adoption but innovation too will set in. Then, there will be nothing holding us back.

To begin with, here are five ways in which technology can help 'green'

your home or office.

1 Start with your lights; try LEDs

For a while now, compact fluorescent lamps (CFLs) have been touted to be energy-savers. Light-emitting diodes (LEDs) are now cropping up as a better alternative. The power consumption of LEDs is less than half that of a CFL, and the electronics associated with driving the LEDs is a lot less complicated compared to CFLs. This helps in reducing the power-line noise. Power supply controllers that drive the LEDs help with power factor (PF) greater than 0.95, resulting in lower total harmonic distortion (THD). These LED lamps are retrofit into the existing sockets.

"Let me show you the cost differential involved in lighting: An 18-watt CFL costs Rs 220 today. An equivalent 6- to 8-watt LED lamp costs Rs 600. However, CFL has 10,000 hours of life compared to 50,000-hour life of an LED lamp. The price of LED lamps will also come down as the volumes ramp up. The reliability of the electronics will only go up, giving everyone an opportunity to adopt LEDs. Rework is nil and the power saving is enormous. Most of all, it is the right thing to do for our planet," urges Ananthaswamy.

The next step, especially for commercial developments, is to go for holistic lighting systems. These can be

designed to accurately control voltage and current regulation for precise LED light intensity and colour mixing, temperature monitoring to prevent thermal runaway, intelligent and adaptive dimming of the LED, and fault detection (over-voltage/over-current, blown-string). Communication with external systems is also possible via power-line communication, wireless technology or interfaces.

Add lighting timers to this, and lights and utilities will automatically dim or turn off after office hours. Public areas can even use motion-sensor-based lighting systems, which can help to reduce power consumption when coupled with the dimming option available with LED lighting.

2 Go for 'green' electronics

Homes can go for Energy Star rated appliances, energy-efficient computers, routers and other 'green' electronics. All this is not at all difficult to find now, because the entire electronics industry is painting its brochures green. Energy Star rated monitors, televisions, refrigerators, microwave ovens, etc are easily available. So are energy-efficient computers with processors that automatically shut down when not in use, lower standby power consumption, LED-backlit displays that are cooler and use less power, etc.



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